

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
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Section / Batch:	B-TECH AIDS	Date:	22/07/2026

Code of Conduct

I JEEVA RAGHUL .M / 192524487 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

```
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int n, int key){
2     int low = 0, high = n-1;
3     while (low<high){
4         int mid = (low + high)/2;
5         if(arr[mid] == key)
6             return mid;
7         else if (arr[mid]<key)
8             low=mid;
9         else
10            high=mid;
11     }
12     return -1;
13 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {
2     if(top==0) {
3         printf("Underflow\n");
4         return -1;
5     }
6     return stack[top--];
7 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3
4 void init() {
5     top=-1;
6 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks



Q2
Stack
5 marks



Q3
Stack
5 marks



Q4
Stack
5 marks



Q5
Stack
5 marks



Q6
Linked List
5 marks



Q7
Stack
5 marks



Q8



Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop0;
}
pop0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(ch==' '){
2     while (stack[top] != '('){
3         postfix[j++]=pop ();
4     }
5     pop ();
6 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||
(ch == ']' && stack[top] == '[')) {
    pop();
} else {
    return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((ch=='[' && stack[top] == '[') ||
2   (ch==']' && stack[top] == '[') ||
3   (ch=='[' && stack[top] == '[')) {
4   pop();
5 }else{
6   return 0;
7 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q6 Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head;
2 while (temp != NULL) {
3     printf("%d\n", temp->data);
4     temp = temp->next;
5 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
[← Back](#)
QUESTIONS
Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int POP(int stack[], int *top) {
2     if(*top == -1){
3         printf("Underflow\n");
4         return -1;
5     }else{
6         return stack[(*top)--];
7     }
8 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7**
Stack
5 marks**Q8** Linked List Data Structures

5 marks

Identify the errors in the following code

```
while current != NULL:
    current.next = prev
    prev = current
    current = current.next
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (current != NULL) {
2     next = current->next;
3     current->next = prev;
4     prev = current;
5     current = next;
6 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)[← Back](#)

QUESTIONS

Q1
Binary Search
5 marks**Q2**
Stack
5 marks**Q3**
Stack
5 marks**Q4**
Stack
5 marks**Q5**
Stack
5 marks**Q6**
Linked List
5 marks**Q7****Q9 Linked List**

5 marks

What is the error in the below code?

new.next = head

head = new

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new->next = head;  
2 head = new;
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS
Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):
if front == -1:
    print("Underflow")
else:
    return Q[front++]
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):
2 if front == -1:
3     print("Underflow")
4 else:
5     item = Q[front]
6     if front == rear:
7         front = rear = -1
8     else:
9         front = front + 1
10    return item
```

Test Input
 stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192524487
RB_6a794db78b7456.58746111\main.c
:1:1: warning: return type
```

Q3
Binary Search
5 marks



Q4
Queue
5 marks



Q5
Queue
5 marks



Q6
Doubly Linked List
5 marks



Q7
Stack
5 marks



Q8
Stack
5 marks



Q9
Circular Queue
5 marks



Q10
Stack



```
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while (expression[i] != '\0'){
2     if (isdigit(expression[i]))
3         push(expression[i] - '0');
4     else{
5         int op1 = pop();
6         int op2 = pop();
7         int result;
8
9         switch (expression[i]){
10            case '+': result = op2 + op1; break;
11            case '-': result = op2 - op1; break;
12            case '*': result = op2 * op1; break;
13            case '/': result = op2 / op1; break;
14        }
15        push(result);
16    }
17    i++;
18 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8

Q2 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {
2     if (front == -1) {
3         printf("Underflow\n");
4         return -1;
5     }
6     int data = queue[front];
7     if (front == rear) {
8         front = rear = -1;
9     } else {
10        front = (front + 1) % MAX;
11    }
12    return data;
13 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack
5 marks

Q8 ✓

Q3 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {
    int mid = l + (r - l) / 2;
    if (arr[mid] == x) return mid;
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);
    return binarySearch(arr, mid, r, x);
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x){
2     if (r >= l){
3         int mid = l + (r - l)/2;
4         if (arr[mid] == x)
5             return mid;
6         if (arr[mid]>x)
7             return binarySearch(arr, l,mid-1,x);
8         return binarySearch(arr,mid+1,r,x);
9     }
10    return -1;
11 }
```

Test Input

stdin input (optional)

Output

QUESTIONS
Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q8
Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
    if (front == 0) front = MAX - 1;
    else front = front - 1;
    deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data){
2     if ((front == 0 && rear == MAX - 1) || (front == rear + 1)){
3         printf("Overflow\n");
4         return;
5     }
6     if (front == -1){
7         front = rear = 0;
8     }else if (front == 0) {
9         front = MAX - 1;
10    }else {
11        front = front -1;
12    }
13    deque[front] = data;
14 }
```

Test Input
 stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data){
2     if (rear == MAX - 1){
3         printf("Overflow\n");
4         return;
5     }
6     if (front == -1){
7         front = 0;
8     }
9     queue[++rear] = data;
10 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack
5 marks

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {
2     struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
3     newNode->data = data;
4     newNode->prev = NULL;
5     newNode->next = head;
6
7     if (head != NULL)
8         head->prev = newNode;
9     head = newNode;
10 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.

POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int POP() {
2     if (top == -1) {
3         printf("Underflow\n");
4         return -1;
5     } else {
6         int item = stack[top];
7         top--;
8         return item;
9     }
10 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)
[← Back](#)
QUESTIONS
Q1 ✓
Stack Notations - Postfix
Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void PUSH(int data){
2     if (top == MAX - 1){
3         printf("Overflow\n");
4         return;
5     }else{
6         stack[++top] = data;
7     }
8 }
```

Test Input
 stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7
Stack
5 marks

Q9 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
    printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int vehicle){
2     if ((front == (rear + 1) % MAX)){
3         printf("Queue Full\n");
4         return;
5     }
6     if (front == -1){
7         front = rear = 0;
8     }else{
9         rear = (rear + 1) % MAX;
10    }
11    queue[rear] = vehicle;
12 }
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1 ✓
Stack Notations - Postfix Expression
5 marks

Q2 ✓
Circular Queue
5 marks

Q3 ✓
Binary Search
5 marks

Q4 ✓
Queue
5 marks

Q5 ✓
Queue
5 marks

Q6 ✓
Doubly Linked List
5 marks

Q7 ✓
Stack

Q10 Stack

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

✳ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int PEEK() {
2     if (top == -1){
3         printf("Stack is empty\n");
4         return -1;
5     }else{
6         return stack[top];
7     }
8 }
```

Test Input

stdin input (optional)

Output